

Abdullah Ahmed Al-Mekhyal

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PROFILE

Senior Computer Science student and Research Assistant at Kuwait University's DATA Lab with applied experience in Artificial Intelligence, Machine Learning, Anti-Money Laundering (AML), graph neural networks, Natural Language Processing, and banking technology. Published IEEE researcher with industry experience at KFH Capital developing enterprise systems and supporting digital transformation. Interested in AI-driven financial crime detection, transaction monitoring, risk analytics, compliance technology, and secure intelligent systems for banks and financial institutions.

EDUCATION

Kuwait University

Bachelor of Computer Science

Sep 2023 – Present

Kuwait

- Coursework: Machine Learning · Data Mining · Natural Language Processing · Database Systems · Algorithms · Web Programming · Statistics

EXPERIENCE

Kuwait University

Research Assistant — DATA Lab

2026 – Present

Kuwait

- Research heterogeneous graph and hypergraph machine learning methods for AML, illicit wallet detection, suspicious transaction analysis, and financial crime analytics.
- Develop reproducible AI experiments using Python, PyTorch Geometric, Scikit-learn, imbalanced-class metrics, and large public financial datasets.

KFH Capital

Technology & Digital Transformation Intern

Jun – Jul 2026

Kuwait

- Contributed to the modernization of internal enterprise applications in an investment-company environment.
- Led development of an internal Reference Management System with a four-member team using React, Node.js, SQL, REST APIs, and role-based access control.
- Collaborated with business stakeholders and technical teams to improve document workflows, reference tracking, usability, and operational efficiency.

Global Student Research (GSR) — KFUPM

Local Ambassador

2025 – Present

GCC

- Represent Kuwait University in regional research initiatives and support collaboration in AI, fintech, data science, and applied innovation.

Computer Science Club — Kuwait University

Secretary

Sep 2024 – Jun 2025

Kuwait

- Coordinated workshops, hackathons, and industry engagement with organizations including Kuwait Finance House and KNPC.

Kuwait National Petroleum Company

IT Intern — Training Center

Jul – Aug 2024

Kuwait

- Gained exposure to enterprise IT systems, information management, data governance, and compliance documentation.

PUBLICATIONS

- “**Heterogeneous Graph Modeling for Anti-Money Laundering: A Reproducible Study on Elliptic++.**” Almekhyal, A. & Joudeh, Y. Kuwait University DATA Lab. ROC-AUC 0.9718, PR-AUC 0.8289, illicit-class recall 0.92.
- “**Handling Code-Switching in Arabic-English Tasks: Benchmarking LLMs on Arabizi and Mixed Arabic-English Text.**” Joudeh, Y. et al. Kuwait University. Benchmarked GPT, Claude, Gemini, and LLaMA using zero-shot and few-shot prompting.
- “**ML-Driven Technique for UAV Threat Detection.**” Al-Mekhyal, A. et al. Published on IEEE Xplore; presented at IEEE SM 2026. Gold Medal, GCC Cybersecurity Summit, Qatar.

SELECTED PROJECTS

Machine Learning-Based Suspicious Transaction Detection | Python · Scikit-learn · AML Analytics | 2025/2026

- Built a reproducible AML classification pipeline on a stratified 950,485-transaction SAML-D sample using 37 numerical features.
- Compared Logistic Regression, Random Forest, Decision Tree, Naive Bayes, and kNN using precision, recall, F1-score, and confusion matrices, with emphasis on minority-class performance.

Reference Management System — KFH Capital | React · Node.js · SQL · REST APIs | 2026

- Managed development of an internal enterprise system for reference numbers, documents, attachments, and employee

workflows.

Motor Vehicle Collisions Analytics Database — NYC | Oracle SQL · Database Design | 2025/2026

- Designed a 3NF relational schema and implemented analytical SQL using joins, window functions, aggregations, views, and structured reporting queries.

Tadreeb — Internship Management Platform | MERN Stack · Role-Based Access Control | 2026

- Developed a full-stack platform connecting students and employers with authentication, opportunity management, application tracking, and administrative approval workflows.

AWARDS & RECOGNITION

Gold Medal — GCC Cybersecurity Summit, Qatar · Google Intelligent Planet Hackathon Finalist · MILSET Expo-Sciences International Representative, Abu Dhabi · Kuwait University “We Proud of You” Recognition

SKILLS & LANGUAGES

Programming	Python, SQL, JavaScript, C, C++
AI & Data	Machine Learning, Graph Neural Networks, Heterogeneous GNN, HMPNN, PyTorch Geometric, Scikit-learn, Data Mining, NLP, LLM Evaluation, Prompt Engineering
Financial Crime	Anti-Money Laundering (AML), Suspicious Transaction Detection, Transaction Monitoring, Fraud Detection, KYC, Risk Analytics, FATF Fundamentals
Web & Software	React, Node.js, Express.js, REST APIs, MERN Stack, Role-Based Access Control, Enterprise Applications
Databases & Tools	Oracle SQL, MongoDB, ER Modelling, Normalisation (1NF–3NF), Git, Linux, Google Colab
Languages	Arabic (Native), English (Professional Proficiency)